

**BARRHEAD, AB, Canada**WMO: **710610**Lat: **54.095N**Lon: **114.448W**Elev: **648**StdP: **93.78**Time Zone: **-7.00 (NAM)**Period: **00-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 12            | -32.9      | -28.8 | -36.4                         | 0.1 | -32.8 | -32.0 | 0.2 | -28.5 | 9.1                   | 2.3  | 7.5 | -2.0 | 1.1                   | 250  | 0.634 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 13.0                   | 28.6            | 17.8 | 26.8 | 16.9 | 25.1 | 16.0 | 19.3                | 26.3 | 18.1 | 24.8 | 17.0 | 23.3 | 3.1                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.7                            | 12.9 | 22.6 | 15.5 | 11.9 | 21.3 | 14.4 | 11.1 | 19.9 | 58.0          | 26.1 | 53.8 | 24.8 | 50.3 | 23.4 | 23.3              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |            |            | Extreme Annual Temperature |             |                    |            | n-Year Return Period Values of Extreme Temperature |             |              |             |              |             |              |             |  |  |
|-------------------|------------|------------|----------------------------|-------------|--------------------|------------|----------------------------------------------------|-------------|--------------|-------------|--------------|-------------|--------------|-------------|--|--|
|                   |            |            | Mean                       |             | Standard Deviation |            | n=5 years                                          |             | n=10 years   |             | n=20 years   |             | n=50 years   |             |  |  |
| 1%                | 2.5%       | 5%         | Min                        | Max         | Min                | Max        | Min                                                | Max         | Min          | Max         | Min          | Max         | Min          | Max         |  |  |
| (a)               | (b)        | (c)        | (e)                        | (f)         | (g)                | (h)        | (i)                                                | (j)         | (k)          | (l)         | (m)          | (n)         | (o)          | (p)         |  |  |
| (4) <b>8.3</b>    | <b>7.0</b> | <b>6.0</b> | <b>-38.9</b>               | <b>31.7</b> | <b>4.7</b>         | <b>1.8</b> | <b>-42.3</b>                                       | <b>33.0</b> | <b>-45.0</b> | <b>34.1</b> | <b>-47.7</b> | <b>35.1</b> | <b>-51.1</b> | <b>36.4</b> |  |  |
| (5)               |            |            | <b>-38.6</b>               | <b>21.2</b> | <b>4.9</b>         | <b>1.1</b> | <b>-42.1</b>                                       | <b>22.0</b> | <b>-44.9</b> | <b>22.7</b> | <b>-47.7</b> | <b>23.3</b> | <b>-51.2</b> | <b>24.1</b> |  |  |

**Monthly Climatic Design Conditions**

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 2.9   | -10.8 | -10.0 | -4.6  | 3.8   | 10.7  | 14.7  | 17.1  | 15.5  | 10.5  | 3.8   | -4.4  | -11.9 | (6)  |
| (7)  |                                                                                  | DBStd       | 12.01 | 8.84  | 8.07  | 7.88  | 5.20  | 4.49  | 3.00  | 2.83  | 3.08  | 4.13  | 4.71  | 6.91  | 7.75  | (7)  |
| (8)  |                                                                                  | HDD10.0     | 3252  | 644   | 561   | 454   | 191   | 46    | 2     | 0     | 2     | 42    | 197   | 433   | 681   | (8)  |
| (9)  |                                                                                  | HDD18.3     | 5657  | 902   | 794   | 712   | 435   | 238   | 115   | 58    | 96    | 235   | 450   | 683   | 939   | (9)  |
| (10) |                                                                                  | CDD10.0     | 672   | 0     | 0     | 0     | 6     | 67    | 143   | 221   | 172   | 58    | 5     | 0     | 0     | (10) |
| (11) |                                                                                  | CDD18.3     | 35    | 0     | 0     | 0     | 0     | 1     | 6     | 19    | 8     | 1     | 0     | 0     | 0     | (11) |
| (12) |                                                                                  | CDH23.3     | 787   | 0     | 0     | 0     | 3     | 66    | 134   | 330   | 189   | 62    | 4     | 0     | 0     | (12) |
| (13) |                                                                                  | CDH26.7     | 169   | 0     | 0     | 0     | 0     | 12    | 26    | 82    | 36    | 13    | 1     | 0     | 0     | (13) |
| (14) | Wind                                                                             | WSAvg       | 2.5   | 2.2   | 2.2   | 2.5   | 3.1   | 3.2   | 2.7   | 2.5   | 2.1   | 2.4   | 2.6   | 2.2   | 2.0   | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 494   | 24    | 15    | 15    | 28    | 49    | 87    | 94    | 64    | 39    | 22    | 22    | 18    | (15) |
| (16) |                                                                                  | PrecMax     | 676   | 106   | 31    | 28    | 65    | 105   | 198   | 200   | 170   | 101   | 62    | 71    | 48    | (16) |
| (17) |                                                                                  | PrecMin     | 340   | 1     | 6     | 4     | 5     | 9     | 43    | 49    | 29    | 6     | 3     | 6     | 2     | (17) |
| (18) |                                                                                  | PrecStd     | 86    | 22    | 8     | 8     | 15    | 24    | 33    | 38    | 28    | 21    | 16    | 16    | 11    | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 8.3   | 9.1   | 12.5  | 23.2  | 28.3  | 29.4  | 31.5  | 30.0  | 28.7  | 22.8  | 12.1  | 7.3   | (19) |
| (20) |                                                                                  |             | MCWB  | 3.3   | 3.7   | 6.0   | 11.4  | 13.9  | 17.2  | 19.8  | 18.4  | 16.7  | 12.6  | 6.1   | 3.4   | (20) |
| (21) |                                                                                  | 2%          | DB    | 5.7   | 5.9   | 9.7   | 18.9  | 25.1  | 26.8  | 28.9  | 27.4  | 24.9  | 16.8  | 8.4   | 4.1   | (21) |
| (22) |                                                                                  |             | MCWB  | 2.3   | 1.7   | 4.0   | 9.3   | 12.8  | 15.9  | 18.9  | 17.9  | 15.2  | 9.7   | 3.8   | 1.0   | (22) |
| (23) |                                                                                  | 5%          | DB    | 3.6   | 3.6   | 7.5   | 15.7  | 22.7  | 24.6  | 26.9  | 25.4  | 21.8  | 14.0  | 5.9   | 1.7   | (23) |
| (24) |                                                                                  |             | MCWB  | 0.7   | 0.4   | 2.8   | 7.3   | 11.7  | 14.8  | 18.0  | 17.0  | 14.0  | 7.6   | 2.3   | -0.5  | (24) |
| (25) |                                                                                  | 10%         | DB    | 1.3   | 1.4   | 5.2   | 13.1  | 20.4  | 22.5  | 24.9  | 23.5  | 18.8  | 11.4  | 3.8   | -0.7  | (25) |
| (26) |                                                                                  |             | MCWB  | -0.8  | -1.0  | 1.5   | 6.0   | 10.8  | 13.9  | 16.8  | 16.3  | 12.3  | 6.4   | 1.3   | -2.4  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 3.7   | 4.0   | 6.2   | 11.6  | 15.0  | 18.7  | 21.6  | 20.3  | 17.5  | 12.8  | 6.5   | 3.6   | (27) |
| (28) |                                                                                  |             | MCDB  | 7.4   | 8.6   | 11.8  | 22.2  | 25.7  | 26.3  | 28.6  | 27.1  | 26.7  | 21.8  | 11.4  | 6.9   | (28) |
| (29) |                                                                                  | 2%          | WB    | 2.3   | 2.1   | 4.4   | 9.6   | 13.6  | 17.0  | 20.0  | 18.9  | 15.8  | 10.0  | 4.2   | 1.3   | (29) |
| (30) |                                                                                  |             | MCDB  | 5.5   | 5.5   | 9.0   | 17.9  | 22.8  | 24.5  | 26.7  | 25.6  | 23.5  | 15.3  | 7.6   | 3.6   | (30) |
| (31) |                                                                                  | 5%          | WB    | 0.9   | 0.6   | 3.0   | 7.8   | 12.5  | 15.8  | 18.6  | 17.7  | 14.2  | 8.2   | 2.7   | -0.4  | (31) |
| (32) |                                                                                  |             | MCDB  | 3.3   | 3.4   | 7.0   | 15.0  | 21.0  | 22.8  | 25.4  | 24.0  | 21.0  | 12.8  | 5.3   | 1.6   | (32) |
| (33) |                                                                                  | 10%         | WB    | -0.8  | -0.9  | 1.7   | 6.3   | 11.3  | 14.8  | 17.5  | 16.7  | 12.8  | 6.8   | 1.5   | -2.4  | (33) |
| (34) |                                                                                  |             | MCDB  | 1.3   | 1.5   | 4.9   | 12.1  | 19.1  | 20.8  | 23.6  | 22.5  | 17.7  | 10.7  | 3.7   | -0.7  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               |             | MDBR  | 11.0  | 12.7  | 12.5  | 12.5  | 13.9  | 12.6  | 13.0  | 13.4  | 13.5  | 11.6  | 9.6   | 10.9  | (35) |
| (36) |                                                                                  | 5% DB       | MCDBR | 12.2  | 14.7  | 14.3  | 18.1  | 18.5  | 17.0  | 17.2  | 17.4  | 19.4  | 16.7  | 12.0  | 12.2  | (36) |
| (37) |                                                                                  |             | MCWBR | 9.2   | 11.1  | 9.3   | 9.6   | 8.4   | 7.7   | 8.2   | 8.9   | 10.2  | 9.7   | 8.1   | 9.8   | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 12.0  | 14.2  | 13.6  | 17.5  | 16.6  | 14.7  | 15.7  | 15.9  | 18.2  | 15.6  | 11.1  | 11.8  | (38) |
| (39) |                                                                                  |             | MCWBR | 9.4   | 10.8  | 9.0   | 9.5   | 7.7   | 7.0   | 8.0   | 8.6   | 9.7   | 9.4   | 7.9   | 9.7   | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.249 | 0.269 | 0.300 | 0.330 | 0.347 | 0.357 | 0.359 | 0.361 | 0.314 | 0.283 | 0.254 | 0.239 | (40)  |      |
| (41) |                                                                                  | taud        | 2.324 | 2.337 | 2.347 | 2.380 | 2.379 | 2.362 | 2.411 | 2.355 | 2.494 | 2.555 | 2.454 | 2.292 | (41)  |      |
| (42) |                                                                                  | Ebn at Noon | 736   | 840   | 885   | 900   | 899   | 890   | 881   | 856   | 865   | 826   | 737   | 667   | (42)  |      |
| (43) |                                                                                  | Edh at Noon | 54    | 75    | 94    | 105   | 111   | 115   | 107   | 107   | 82    | 61    | 48    | 45    | (43)  |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.89  | 1.89  | 3.34  | 4.61  | 5.55  | 5.85  | 5.92  | 4.92  | 3.40  | 1.93  | 0.93  | 0.64  | (44)  |      |
| (45) |                                                                                  | RadStd      | 0.07  | 0.14  | 0.25  | 0.41  | 0.43  | 0.41  | 0.37  | 0.36  | 0.35  | 0.21  | 0.09  | 0.06  | (45)  |      |

**Historical Trends**

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (49 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page